Solvency Field Theory V10.2: Projection Geometry Closure

The $4\pi^2$ Spine, the Unified Screen Family, and the SPARC Morphology Confirmation

A standalone extension paper building on V10 and V10.1

April 2026

Purpose. V10 established the structural backbone of Solvency Field Theory. V10.1 pushed the early-universe shatter program to the level of field-theoretic instability, Boltzmann insertion, CLASS implementation, calibration, and low-ℓ phenomenology. The present paper preserves the next closure cluster. Three theorems close in sequence and they are not independent results. C19 derives $\epsilon_\Phi = 1/(4\pi^2)$ from a minimal Onsager interface law, completing the fifth appearance of the $4\pi^2$ structural spine. Theorem 5 shows that the cosmological softening factors (1/3, 2/3) and the galactic response coefficients (4/3, $3\pi/8$) are members of one projection family on S^2. Theorem 4 then confirms the resulting morphology dependence at the v1 empirical level in SPARC when morphology is parameterized by Hubble type and non-morphological nuisance control is handled correctly. The chain runs: Onsager closes the geometric spine → the projection family generates the morphology curve → SPARC confirms it.

Abstract

This extension preserves three linked advances in Solvency Field Theory (SFT). First, we close the remaining phase-convention slot in the shatter-radiation transfer law by deriving

$$\epsilon_\Phi = \frac{1}{4\pi^2}$$

from a minimal near-equilibrium Onsager interface law. The derivation makes the two cycle-angle Jacobians explicit and cleanly separates bookkeeping normalization from the genuine transport block. Second, we show that the cosmological projection coefficients

$$s_{\text{wall}} = \frac{1}{3}, \qquad s_{\text{filament}} = \frac{2}{3}$$

and the galactic coefficients

$$\lambda_{\text{sph}} = \frac{4}{3}, \qquad \lambda_{\text{disk}} = \frac{3\pi}{8}$$

belong to one family

$$M(w, P; c) = c\, \langle w(\theta) P(\theta) \rangle_{S^2} ,$$

where source geometry fixes the participation weight w, the observable fixes the projection kernel P, and the sector equation fixes the prefactor c. This yields a continuous Gamma-family morphology

curve

$$\lambda(\alpha) = \sqrt{\pi}\,\frac{\Gamma((\alpha+4)/2)}{\Gamma((\alpha+5)/2)}$$

interpolating from isotropic sphere to ideal thin disk. Third, we freeze the empirical closure of the morphology theorem at the v1 level. In 131 interior SPARC galaxies, Hubble type T provides the correct morphology coordinate; the rank correlation with best-fit λ_{obs} is negative and highly significant,

$$\rho(T, \lambda_{\text{obs}}) = -0.3005, \qquad p = 4.87 \times 10^{-4},$$

and remains significant under inclination control,

$$\rho(T, \lambda_{\text{obs}} \mid \text{inclination}) = -0.2732, \qquad p = 0.0016.$$

The early-type endpoint lands at median $\lambda = 1.340$ against the spherical target $4/3 = 1.333$. The honest status after this paper is therefore: the phase-convention slot in Theorem 3 is closed; the unified screen family of Theorem 5 is closed; and Theorem 4 is closed at the v1 empirical level, while a sharper direct map from structural photometry to the theory parameter α remains a later refinement target.

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1 Why V10.2 exists

V10 did two things at once. It made a bold framework-level claim — that expansion and gravity are two geometric shapes of one information-maintenance process — and it also started enforcing discipline about what was derived, what was numerically supported, what was still candidate status, and what remained open. V10.1 then did the same thing for the early-universe sector: it took the shatter program from a suggestive candidate architecture to a solver-facing pipeline. That left a very specific kind of work behind. Not another monolithic rewrite. Not another broad synthesis. A closure cluster.

This paper preserves that cluster.

The first piece is the last unclosed symbolic slot in the shatter-radiation transfer law. Before C19, the transfer coefficient had already been factorized into thermodynamic conversion, baryon/radiation partition factors, a transport block, and a remaining phase-convention slot ϵ_Φ. The structure was already visible. What was missing was the reason the interface should carry the specific factor $1/(4\pi^2)$ without hiding it inside transport microphysics.

The second piece is the projection story. In V10 the cosmological and galactic sectors already exhibited structurally parallel coefficients, but the parallel had not yet been written as one theorem-level family. Theorem 5 does that. It makes explicit that the wall, filament, sphere, and disk coefficients are not separate coincidences; they are evaluations of one spherical projection family with different participation weights, kernels, and sector prefactors.

The third piece is empirical. Once the Gamma-family morphology curve exists on the theory side, the framework owes the reader an honest answer to a simple question: does the data move in that direction when morphology is parameterized correctly? The first pass at that question used a crude bulge-fraction proxy. That proxy was good enough to show endpoint splits, but not good enough to serve as the continuous morphology coordinate. The correct coordinate for the v1 closure pass is SPARC Hubble type T, and once that choice is made, the signal survives the right nuisance audit.

That is why V10.2 exists. It is not another beginning. It is the preservation of three specific closures that now belong together.

2 The structural chain in one line

The argument of this paper is short enough to say in one sentence and strong enough that it should be said plainly:

> The fifth and final appearance of the $4\pi^2$ spine closes because the shatter-radiation interface is dissipative and therefore Onsager, not ad hoc; the same projection geometry that fixes the wall, filament, sphere, and disk coefficients generates the continuous morphology curve; and SPARC confirms that curve once morphology is parameterized by the right variable.

The sections that follow unpack that sentence. They also keep the status boundaries clean.

- Section 3 closes the phase-convention interface factor in Theorem 3.
- Section 4 proves the unified screen family of Theorem 5.
- Section 5 freezes the v1 empirical closure of Theorem 4.
- Section 6 states exactly what is now closed and what remains a later refinement target.

3 Theorem 3 update: Onsager closes the phase slot

3.1 What remained open after the earlier shatter program

After the factorization work in the shatter transfer chain, the effective source law had the form

$$\delta Q_{\gamma,k} = q_T f_b f_r \, \epsilon_\Phi \, (\mathcal{R}_\Phi K_\Phi) \, \frac{k^2}{a^2} \, \delta\Phi_k.$$

By that stage the unresolved question was not whether a source existed. It was much narrower and much cleaner: why should the phase-convention interface factor be exactly $1/(4\pi^2)$?

That matters because the $4\pi^2$ recurrence is not decoration in SFT. It is the same structural factor that already appears in the modified Friedmann ratio and in the weak-field scale. If the CMB appearance were only a numerical splice, the geometric spine would remain suggestive but incomplete. If it can be derived at the interface level, the spine becomes five appearances deep.

3.2 Why the interface must be dissipative

The shatter-radiation boundary is not a conservative matching problem. It is an interface where a broken-phase ledger variable communicates with a radiation mode through heating. That is a dissipative statement. The correct formal language is therefore not a conservative projection trick but linear nonequilibrium thermodynamics.

The source side and sink side are written in different natural conventions:

- SFT counts irreversible maintenance in complete cycles.
- Canonical field theory and canonical radiation modes are naturally angular.

That mismatch is the whole point. If the interface law is written in the correct thermodynamic variables, the cycle-angle Jacobians have to appear explicitly somewhere. The question is whether they are real interface normalizations or whether they should be hidden inside transport. Minimal Onsager closure answers that.

3.3 The two unavoidable Jacobians

Let the source-side order variable be written in cycles as Φ and define the canonically angular field

$$\theta = 2\pi\Phi.$$

If the free energy is $F[\Phi]$, then the cycle-based and angular chemical potentials are related by the chain rule:

$$\mu_\Phi \equiv \frac{\delta F}{\delta \Phi}, \qquad \mu_\theta \equiv \frac{\delta F}{\delta \theta}, \qquad \mu_\Phi = 2\pi \mu_\theta.$$

Therefore

$$\mu_\theta = \frac{\mu_\Phi}{2\pi}.$$

That is the first unavoidable factor of $1/(2\pi)$.

Now let the photon-baryon mode be written in an angular phase variable ϑ_γ, while the SFT bookkeeping counts cycle advancement:

$$N_\gamma = \frac{\vartheta_\gamma}{2\pi}, \qquad \dot{N}_\gamma = \frac{\dot{\vartheta}_\gamma}{2\pi}.$$

That is the second independent factor of $1/(2\pi)$.

The important point is that these are not transport assumptions. They are variable normalizations forced by how the two sides of the interface are written.

3.4 Minimal Onsager closure

Define the source-side thermodynamic force by

$$X_\Phi \equiv \frac{\mu_\theta}{T} = \frac{1}{2\pi}\frac{\mu_\Phi}{T},$$

and the sink-side flux by the radiation cycle current

$$J_\gamma \equiv \dot{N}_\gamma = \frac{\dot{\vartheta}_\gamma}{2\pi}.$$

The minimal linear-response law is then

$$J_\gamma = L_{\gamma\Phi} X_\Phi, \qquad L_{\gamma\Phi} > 0.$$

The entropy production density is

$$\sigma = J_\gamma X_\Phi = L_{\gamma\Phi} X_\Phi^2 \geq 0.$$

So the interface law is not just plausible. It is the simplest admissible dissipative closure.

Before specifying any transport detail, the dissipated power density is

$$\dot{q}_{\text{diss}} = T\sigma = T J_\gamma X_\Phi.$$

Substituting the cycle-angle conversions gives

$$\dot{q}_{\text{diss}} = T\left(\frac{\dot{\vartheta}_\gamma}{2\pi}\right)\left(\frac{\mu_\Phi}{2\pi T}\right) = \frac{1}{4\pi^2}\mu_\Phi \dot{\vartheta}_\gamma.$$

So the interface itself carries

$$\boxed{\epsilon_\Phi = \frac{1}{4\pi^2}.}$$

That is the exact point of closure. One factor comes from converting the source-side ledger variable from cycles to a canonical angle. One comes from converting the sink-side radiation mode from a canonical angle to cycles. Neither belongs in the transport coefficient.

3.5 Recovery of the source law

The observable Boltzmann source is linear in $\delta\Phi_k$, not bilinear in μ_Φ and $\dot{\vartheta}_\gamma$, so we expand around a nonzero background radiation cycle rate:

$$\dot{\vartheta}_\gamma = \dot{\bar{\vartheta}}_\gamma + \delta\dot{\vartheta}_\gamma, \qquad \mu_\Phi = \bar{\mu}_\Phi + \delta\mu_\Phi.$$

At the background broken minimum,

$$\bar{\mu}_\Phi = 0,$$

so the first-order heating rate is

$$\delta\dot{q}_{\gamma,k} = q_T f_b f_r \, \frac{\dot{\bar{\vartheta}}_\gamma}{4\pi^2} \, \delta\mu_{\Phi,k}.$$

Near the interface, the shatter chemical potential is dominated by the gradient term:

$$\delta\mu_\Phi \approx -\frac{K_\Phi}{a^2}\nabla^2\delta\Phi.$$

In Fourier space,

$$\delta\mu_{\Phi,k} \approx K_\Phi \frac{k^2}{a^2}\delta\Phi_k.$$

Insert this into the linearized law:

$$\delta\dot{q}_{\gamma,k} = q_T f_b f_r \, \frac{1}{4\pi^2}(\dot{\bar{\vartheta}}_\gamma K_\Phi)\frac{k^2}{a^2}\delta\Phi_k.$$

Thus

$$\beta_Q(\eta) = q_T f_b(\eta) f_r(\eta) \, \frac{\mathcal{R}_\Phi(\eta) K_\Phi(\eta)}{4\pi^2}, \qquad \mathcal{R}_\Phi(\eta) \equiv \dot{\bar{\vartheta}}_\gamma(\eta),$$

and the source recovers the expected form

$$\delta Q_{\gamma,k} = \beta_Q(\eta)\frac{k^2}{a^2}\delta\Phi_k.$$

3.6 What is now closed

The honest status after this derivation is straightforward.

- The phase-convention interface factor is derived.
- The k^2/a^2 source structure is derived from the gradient chemical potential.

- The transport block $\mathcal{R}_\Phi(\eta)K_\Phi(\eta)$ remains transport physics.

That is a good closure. It narrows the remaining work to exactly what actually remains: microphysical transport refinement and eventual table-driven or microscopic solver input. The bookkeeping factor is no longer open.

4 Theorem 5: one projection family, four coefficients

4.1 Why the projection story needed its own theorem

By V10, SFT already had four coefficients that looked too structurally aligned to be independent accidents:

$$s_{\text{wall}} = \frac{1}{3}, \qquad s_{\text{filament}} = \frac{2}{3}, \qquad \lambda_{\text{sph}} = \frac{4}{3}, \qquad \lambda_{\text{disk}} = \frac{3\pi}{8}.$$

Two lived in the cosmological sector. Two lived in the galactic sector. The temptation was to say they felt related. That is not enough. Structural unification becomes real only when the coefficients are written as members of one family and their differences are shown to come from geometry and sector equation, not from hand-placed constants.

Theorem 5 does exactly that.

4.2 The unified family

Define

$$M(w, P; c) = c\,\langle w(\theta)P(\theta)\rangle_{S^2}\,,$$

where:

- $w(\theta)$ is the participation weight fixed by source geometry,
- $P(\theta)$ is the projection kernel fixed by the observable,
- c is the sector prefactor fixed by the physical equation the projection enters.

The four coefficients then fall into one table:

Structure	$w(\theta)$	$P(\theta)$	c	M	Sector
Wall	1	$\cos^2\theta$	1	$1/3$	Cosmology
Filament	1	$\sin^2\theta$	1	$2/3$	Cosmology
Sphere	1	$\sin^2\theta$	2	$4/3$	Galactic
Disk	$\sin\theta$	$\sin^2\theta$	2	$3\pi/8$	Galactic

Everything changes where it should change. Nothing changes where it should not.

4.3 The spherical moments

The backbone integrals are the moments

$$J_n \equiv \langle \sin^n \theta \rangle_{S^2} = \frac{1}{2} \int_0^\pi \sin^{n+1} \theta \, d\theta.$$

The needed values are

$$J_0 = 1, \qquad J_2 = \frac{2}{3}, \qquad J_3 = \frac{3\pi}{16},$$

and

$$\left\langle \cos^2 \theta \right\rangle_{S^2} = 1 - J_2 = \frac{1}{3}.$$

The cosmological coefficients follow immediately:

$$s_{\text{wall}} = \left\langle \cos^2 \theta \right\rangle_{S^2} = \frac{1}{3}, \qquad s_{\text{filament}} = \left\langle \sin^2 \theta \right\rangle_{S^2} = \frac{2}{3}.$$

The galactic coefficients follow from the same moments but with a different sector prefactor and, in the disk case, a different participation field:

$$\lambda_{\text{sph}} = 2 \left\langle \sin^2 \theta \right\rangle_{S^2} = 2J_2 = \frac{4}{3},$$

$$\lambda_{\text{disk}} = 2 \left\langle \sin \theta \, \sin^2 \theta \right\rangle_{S^2} = 2J_3 = \frac{3\pi}{8}.$$

The important conceptual point is that the galactic prefactor $c = 2$ is not another geometric ingredient. It comes from the quadratic weak-field response law

$$g_{\text{eff}}^2 = g_{\text{bar}}^2 + \lambda \, a_{\text{SFT}} g_{\text{bar}},$$

while the cosmological sector uses a linear burden correction. The projection itself lives in the integral. The sector equation decides what multiplies it.

4.4 Three structural identities

Once the four coefficients are written this way, three identities are immediate.

Identity 1: completeness

$$s_{\text{wall}} + s_{\text{filament}} = \frac{1}{3} + \frac{2}{3} = 1.$$

This is just

$$\left\langle \cos^2 \theta + \sin^2 \theta \right\rangle_{S^2} = 1.$$

The wall and filament corrections are not separate free quantities. They partition a unit burden field.

Identity 2: scale bridge

$$\lambda_{\text{sph}} = 2\, s_{\text{filament}} \qquad \Longrightarrow \qquad \frac{4}{3} = 2 \times \frac{2}{3}.$$

This is the deepest identity in the theorem. The spherical galactic response coefficient and the cosmological filament softening factor are the same integral J_2 seen through two different sector equations. The galactic and cosmological sectors are therefore measuring the same geometric object at different scales.

Identity 3: morphology bridge

The wall and disk coefficients are not related by one-step algebra because they belong to different moments (J_2 versus J_3), but they still live on the same S^2 kernel family. The bridge between them is not a constant. It is a continuous morphology interpolation.

4.5 The Gamma-family morphology curve

Define the participation family

$$w(\theta; \alpha) = \sin^{\alpha} \theta, \qquad 0 \le \alpha \le 1.$$

This interpolates from isotropic sphere ($\alpha = 0$) to ideal thin disk ($\alpha = 1$).

The resulting morphology curve is

$$\lambda(\alpha) = 2 \left\langle \sin^{\alpha} \theta \, \sin^2 \theta \right\rangle_{S^2} = \sqrt{\pi}\, \frac{\Gamma((\alpha+4)/2)}{\Gamma((\alpha+5)/2)}.$$

It satisfies

$$\lambda(0) = \frac{4}{3}, \qquad \lambda(1) = \frac{3\pi}{8},$$

and decreases monotonically from sphere to disk.

This is the theory-side statement of Theorem 4. The geometry generates a continuous curve. The empirical question is whether real galaxies order themselves along it in the expected direction.

4.6 What Theorem 5 means

Theorem 5 says the framework does not have two projection stories. It has one.

The cosmological ladder and the galactic weak-field response are not independent sectors that happened to pick related integrals. They are two physical applications of one construction on S^2. That matters because it turns the morphology dependence from a local galactic curiosity into part of a larger structural claim. Once the same family generates both the cosmological and galactic coefficients, the morphology curve is no longer an isolated fit. It is the continuation of the same geometry.

5 Theorem 4: SPARC confirms the morphology curve at v1

5.1 What had to be fixed first

The first empirical pass used a bulge-fraction proxy extracted from the mass model. That proxy was useful, but it had a structural limitation: for most galaxies it was exactly zero. It could test endpoint splits. It could not serve as the continuous morphology coordinate the theorem actually wants.

That matters because Theorem 4 is not just an endpoint theorem. The theory side already gives a continuous morphology curve $\lambda(\alpha)$. The empirical side therefore has to use a morphology variable that is genuinely ordered across the sample.

The correct v1 choice is SPARC Hubble type T.

5.2 Profiling procedure

The per-galaxy best-fit solvency coefficient λ_{obs} is obtained by profiling over the stellar mass-to-light nuisance grids:

$$\lambda \in [0.30,\, 2.50], \qquad \Upsilon_{\text{disk}} \in [0.35,\, 0.75], \qquad \Upsilon_{\text{bulge}} \in [0.45,\, 1.05],$$

with $\Upsilon_{\text{bulge}} = 0$ when no bulge component is present.

At each λ, the fit minimizes χ^2 over the $(\Upsilon_{\text{disk}}, \Upsilon_{\text{bulge}})$ grid. The best-fit λ_{obs} is the grid point with minimum profiled χ^2, and the uncertainty σ_λ is taken from the $\Delta\chi^2 = 1$ interval.

After quality cuts and removal of edge-hit cases, the interior sample contains

$$n = 131$$

galaxies with SPARC metadata.

5.3 Why Hubble type is the right morphology variable

SPARC Hubble type T runs from

$$T = 0 \text{ (S0)} \quad \text{to} \quad T = 11 \text{ (BCD)}.$$

It provides a continuous ordered morphology coordinate across the full sample. That is exactly what the empirical side needs.

The bulge-fraction proxy remains useful supporting evidence, but not the closure variable. The reason is simple: it collapses too much of the sample onto the same value. Hubble type does not.

5.4 Primary correlation result

For the 131 interior galaxies, the rank correlation between morphology and best-fit solvency coefficient is

$$\rho(T, \lambda_{\text{obs}}) = -0.3005, \qquad p = 4.87 \times 10^{-4}.$$

The sign is the one the theory predicts:

$$\text{earlier type} \implies \text{more bulge-dominated} \implies \text{higher } \lambda.$$

That is already enough to say the data move in the correct direction. The stronger question is whether the signal survives nuisance control.

5.5 What counts as a nuisance, and what does not

Inclination is a real nuisance. It changes how cleanly the observed velocity field and the baryonic decomposition are measured. Face-on and edge-on systems do not present the same observational problem. So inclination belongs in a control test.

Gas fraction does *not* belong in the same way. Gas richness is entangled with morphology itself. Gas-rich galaxies tend to be disk-dominated; gas-poor galaxies tend to be earlier-type and more bulge-heavy. Controlling for gas fraction therefore removes part of the morphology signal one is trying to test. That is not a neutral nuisance correction. It is a methodological category error.

The right v1 nuisance logic is therefore:

- control non-morphological observational systematics such as inclination,
- do *not* use morphology-linked quantities as if they were external nuisance variables.

5.6 Partial correlation under inclination control

Once inclination is controlled, the signal remains:

$$\rho(T, \lambda_{\text{obs}} \mid \text{inclination}) = -0.2732, \qquad p = 0.001594.$$

A stronger robustness check also controlling the total $3.6\,\mu$m luminosity gives

$$\rho(T, \lambda_{\text{obs}} \mid \text{inclination}, L_{3.6}) = -0.2310, \qquad p = 0.007940.$$

The signal weakens but remains significant at $p < 0.01$. That is exactly the behavior one expects when the second control variable is not purely external to morphology.

5.7 Endpoint evidence

The endpoint evidence points in the same direction.

For early types,

$$T \leq 4 \quad (S0 \text{ through } Sbc),$$

the sample has

$$n = 37, \qquad \text{median } \lambda = 1.3400.$$

The spherical target is

$$\lambda_{\text{sph}} = \frac{4}{3} = 1.3333.$$

So the early-type endpoint lands almost exactly where the sphere-side theorem says it should.

A broad early-vs-late split also survives:

$$\text{Mann-Whitney}(T \leq 4 > T \geq 7): \qquad p = 0.005069.$$

The late side is more heterogeneous because Hubble type extends past the regular spiral sequence into irregular and BCD systems. That is why the main closure claim of this paper does not rest on forcing the whole late tail to coincide with the ideal thin-disk endpoint. The v1 theorem claim is narrower and stronger: once morphology is parameterized correctly, the galaxies order themselves in the predicted direction and the signal survives the proper nuisance audit.

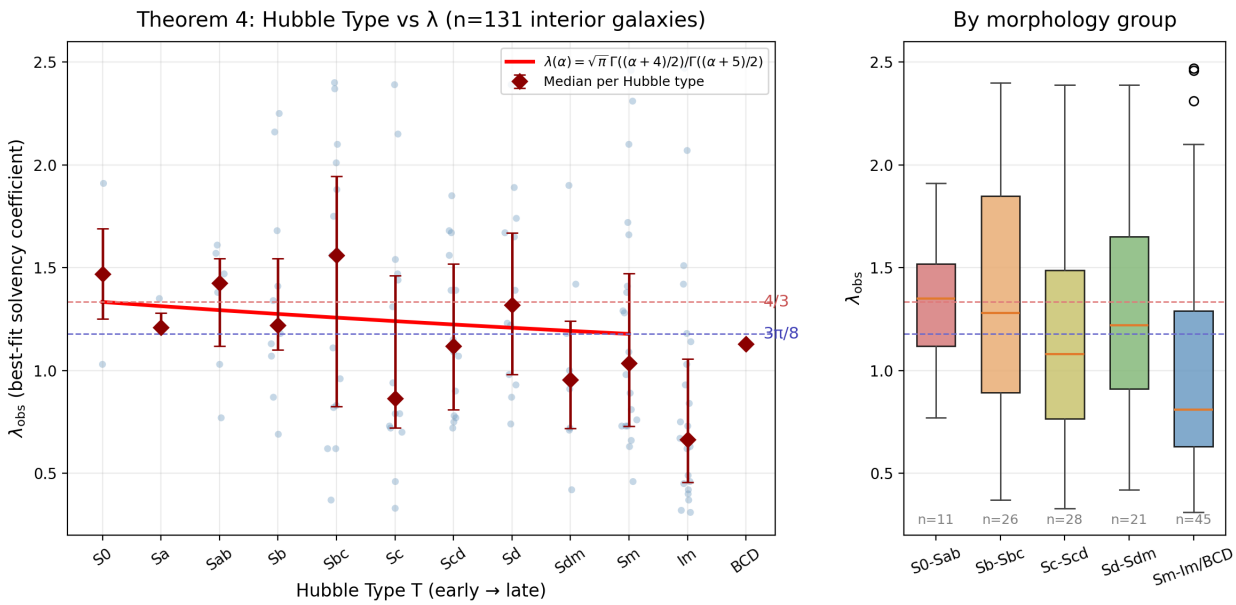

Figure 1: **Theorem 4 closure figure.** Left: interior SPARC galaxies plotted by Hubble type with median λ_{obs} per type and a qualitative Gamma-family overlay. The overlay is shown as a guide to the expected sphere-to-disk direction across the regular sequence; it is not a claimed unique calibration of $\alpha(T)$. Right: grouped morphology bins showing the same ordering at the population level. The empirical closure claim of V10.2 rests on the significant negative rank relation between Hubble type and λ_{obs} under inclination control, not on a unique fitted $T \mapsto \alpha$ map.

5.8 What is closed, and what is not

The v1 empirical closure of Theorem 4 is now:

> When morphology is parameterized by SPARC Hubble type, the SFT morphology dependence is confirmed with the expected sign and with significant persistence under inclination control.

What remains for v2 is a sharper direct observational map from structural photometry — for example B/T, concentration, or Sersic structure — onto the theory parameter α. That is a refinement target, not a blocker for the present theorem status.

6 Updated theorem ledger

The point of V10.2 is not only to present results but also to update the status map cleanly.

Object	What V10/V10.1 left open	Status after V10.2
Theorem 3 / CMB transfer phase slot	ϵ_Φ still carried as the remaining phase-convention interface factor	Closed at symbolic level: $\epsilon_\Phi = 1/(4\pi^2)$ derived from Onsager interface closure; transport block remains refinement target
Theorem 5 / projection unification	Structural parallel visible but not yet promoted to one theorem family	Closed: wall, filament, sphere, and disk coefficients written as one S^2 projection family with explicit scale bridge
Theorem 4 / morphology dependence	Theory-side curve available, empirical closure variable not yet frozen	Closed at v1 empirical level under Hubble-type parameterization with inclination-controlled significance; sharper direct α map remains v2 refinement target

This is the right level of optimism. Nothing in this paper depends on pretending the program is finished in every imaginable sense. But it also does not undersell what is actually earned. Three things have now crossed from "not yet fully preserved" to "preserved and load-bearing."

7 Conclusion

The work preserved in this extension is best understood as one argument with three steps.

First, the fifth appearance of the $4\pi^2$ spine no longer sits at candidate-splice status in the phase slot. The shatter-radiation interface is a dissipative interface, and once it is written in the correct Onsager variables, the two cycle-angle Jacobians become explicit and force

$$\epsilon_\Phi = \frac{1}{4\pi^2}.$$

Second, the projection coefficients scattered across the cosmological and galactic sectors are no longer separate empirical facts waiting for a story. They are members of one family on S^2:

$$M(w, P; c) = c \, \langle w(\theta) P(\theta) \rangle_{S^2} \, .$$

The cosmological wall and filament coefficients and the galactic sphere and disk coefficients are different evaluations of that family, with the same geometric object appearing across scales. The continuous morphology curve is therefore not an add-on. It is the natural continuation of the same projection geometry.

Third, the empirical closure of the morphology theorem now uses the right variable. Once morphology is parameterized by SPARC Hubble type, the signal is clear, correctly signed, and robust under the appropriate nuisance control. That is enough to preserve Theorem 4 at the v1 empirical level.

So the clean summary is:

- The geometric spine is now symbolically complete at five appearances.
- The screen family is now theorem-level rather than suggestive.
- The morphology dependence is now preserved at the v1 empirical level.

That is what V10.2 contributes.

A Appendix A: Full Onsager derivation of ϵ_Φ

This appendix preserves the full derivation in one place so it can later be split cleanly into a companion structure.

A.1 A.1 Remaining source law

The factorized shatter source law is

$$\delta Q_{\gamma,k} = q_T f_b f_r \, \epsilon_\Phi \, (\mathcal{R}_\Phi K_\Phi) \, \frac{k^2}{a^2} \, \delta\Phi_k.$$

All factors except ϵ_Φ had already been isolated structurally.

A.2 A.2 Source-side normalization

With $\theta = 2\pi\Phi$,

$$\mu_\Phi = \frac{\delta F}{\delta \Phi}, \qquad \mu_\theta = \frac{\delta F}{\delta \theta}, \qquad \mu_\Phi = 2\pi\mu_\theta.$$

Hence

$$\mu_\theta = \frac{\mu_\Phi}{2\pi}.$$

A.3 A.3 Sink-side normalization

With radiation phase ϑ_γ and cycle variable $N_\gamma = \vartheta_\gamma/(2\pi)$,

$$\dot{N}_\gamma = \frac{\dot{\vartheta}_\gamma}{2\pi}.$$

A.4 A.4 Onsager variables

Define

$$X_\Phi = \frac{\mu_\theta}{T} = \frac{1}{2\pi}\frac{\mu_\Phi}{T}, \qquad J_\gamma = \dot{N}_\gamma = \frac{\dot{\vartheta}_\gamma}{2\pi}.$$

Minimal linear response gives

$$J_\gamma = L_{\gamma\Phi} X_\Phi, \qquad L_{\gamma\Phi} > 0.$$

Then

$$\sigma = J_\gamma X_\Phi = L_{\gamma\Phi} X_\Phi^2 \geq 0.$$

A.5 A.5 Dissipated power and the phase slot

$$\dot{q}_{\text{diss}} = T\sigma = T J_\gamma X_\Phi = T\left(\frac{\dot{\vartheta}_\gamma}{2\pi}\right)\left(\frac{\mu_\Phi}{2\pi T}\right) = \frac{1}{4\pi^2}\mu_\Phi \dot{\vartheta}_\gamma.$$

So

$$\boxed{\epsilon_\Phi = \frac{1}{4\pi^2}.}$$

A.6 A.6 Linearized source recovery

Write

$$\dot{\vartheta}_\gamma = \dot{\bar{\vartheta}}_\gamma + \delta\dot{\vartheta}_\gamma, \qquad \mu_\Phi = \bar{\mu}_\Phi + \delta\mu_\Phi.$$

At the broken minimum $\bar{\mu}_\Phi = 0$, so

$$\delta\dot{q}_{\gamma,k} = q_T f_b f_r \frac{\dot{\bar{\vartheta}}_\gamma}{4\pi^2}\delta\mu_{\Phi,k}.$$

Using

$$\delta\mu_\Phi \approx -\frac{K_\Phi}{a^2}\nabla^2\delta\Phi \quad \Longrightarrow \quad \delta\mu_{\Phi,k} \approx K_\Phi \frac{k^2}{a^2}\delta\Phi_k,$$

gives

$$\delta\dot{q}_{\gamma,k} = q_T f_b f_r \frac{1}{4\pi^2}(\dot{\bar{\vartheta}}_\gamma K_\Phi)\frac{k^2}{a^2}\delta\Phi_k.$$

Therefore

$$\beta_Q(\eta) = q_T f_b(\eta) f_r(\eta)\frac{\mathcal{R}_\Phi(\eta)K_\Phi(\eta)}{4\pi^2}.$$

B Appendix B: Full projection-family derivation

B.1 B.1 Moments on S^2

$$J_n = \langle \sin^n\theta\rangle_{S^2} = \frac{1}{2}\int_0^\pi \sin^{n+1}\theta\, d\theta.$$

The relevant moments are

$$J_0 = 1, \qquad J_2 = \frac{2}{3}, \qquad J_3 = \frac{3\pi}{16}.$$

B.2 B.2 Wall and filament

$$s_{\text{wall}} = \left\langle \cos^2\theta \right\rangle_{S^2} = 1 - J_2 = \frac{1}{3}, \qquad s_{\text{filament}} = \left\langle \sin^2\theta \right\rangle_{S^2} = J_2 = \frac{2}{3}.$$

B.3 B.3 Sphere and disk

$$\lambda_{\text{sph}} = 2\left\langle \sin^2\theta \right\rangle_{S^2} = 2J_2 = \frac{4}{3}.$$

For the disk,

$$\lambda_{\text{disk}} = 2\left\langle \sin\theta\,\sin^2\theta \right\rangle_{S^2} = 2J_3 = \frac{3\pi}{8}.$$

B.4 B.4 The morphology interpolation

For

$$w(\theta;\alpha) = \sin^\alpha\theta, \qquad 0 \le \alpha \le 1,$$

the response coefficient is

$$\lambda(\alpha) = 2\left\langle \sin^\alpha\theta\,\sin^2\theta \right\rangle_{S^2} = \int_0^\pi \sin^{\alpha+3}\theta\, d\theta.$$

Using

$$\int_0^\pi \sin^m\theta\, d\theta = \sqrt{\pi}\,\frac{\Gamma((m+1)/2)}{\Gamma((m+2)/2)},$$

with $m = \alpha + 3$, gives

$$\boxed{\lambda(\alpha) = \sqrt{\pi}\,\frac{\Gamma((\alpha+4)/2)}{\Gamma((\alpha+5)/2)}.}$$

Hence

$$\lambda(0) = \frac{4}{3}, \qquad \lambda(1) = \frac{3\pi}{8}.$$

C Appendix C: Theorem 4 empirical methods freeze

C.1 C.1 Sample and profiling

Total profiled galaxies:

$$149.$$

Interior galaxies after removing edge-hit cases and requiring SPARC metadata:

$$131.$$

The fit uses

$$\nu(x) = \sqrt{1 + \lambda/x}, \qquad a_{\mathrm{SFT}} = \frac{cH_0}{2\pi} = 1.047 \times 10^{-10}\ \mathrm{m\,s^{-2}},$$

with

$$\lambda \in [0.30, 2.50], \quad \Upsilon_{\mathrm{disk}} \in [0.35, 0.75], \quad \Upsilon_{\mathrm{bulge}} \in [0.45, 1.05].$$

C.2 C.2 Closure variable

The empirical morphology coordinate is SPARC Hubble type T. The bulge-fraction proxy

$$\mathrm{bulge_frac} = \frac{\max|V_{\mathrm{bul}}|}{\max|V_{\mathrm{bul}}| + \max|V_{\mathrm{disk}}|}$$

remains supporting evidence only.

C.3 C.3 Primary result

$$\rho(T, \lambda_{\mathrm{obs}}) = -0.3005, \qquad p = 0.000487.$$

C.4 C.4 Partial correlation

$$\rho(T, \lambda_{\mathrm{obs}} \mid \mathrm{inclination}) = -0.2732, \qquad p = 0.001594.$$

Robustness check:

$$\rho(T, \lambda_{\mathrm{obs}} \mid \mathrm{inclination}, L_{3.6}) = -0.2310, \qquad p = 0.007940.$$

C.5 C.5 Gas fraction is not a nuisance

Gas fraction is morphology-linked, not external to morphology. It should therefore not be used as a nuisance variable in the closure test of a morphology theorem.

C.6 C.6 Endpoint evidence

Early types:

$$T \leq 4, \qquad n = 37, \qquad \text{median}\,\lambda = 1.3400.$$

Spherical target:

$$\frac{4}{3} = 1.3333.$$

Population split:

$$\text{Mann-Whitney}(T \leq 4 > T \geq 7): \quad p = 0.005069.$$

C.7 C.7 Closure statement

At the v1 empirical level, the SFT morphology dependence is confirmed when morphology is parameterized by SPARC Hubble type T, with the expected sign and with significant persistence under inclination control. A sharper direct map from structural photometry to the theory parameter α remains a later refinement target.